A Multi-Gb/s Frame-Interleaved LDPC Decoder With Path-Unrolled Message Passing in 28-nm CMOS

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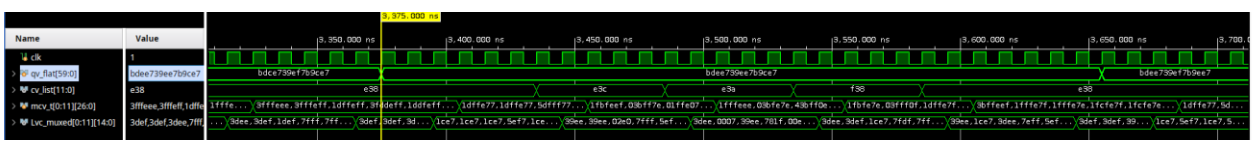
SIMULATION RESULTS



Simulation has been presented for:-

- No. of macrocolumns = 4
- No. Of macrorows = 3
- Size of submatrix = 3 x 3
- MAX_ITERATIONS = 5
- Block Length= No. of macrocolumns x Size of submatrix = 12
- Frames = 1

Cycles per iterations = No. of macrocolumns+2

Latency = Cycles per iterations x MAX_ITERATIONS = 6 x 5 = 30 clock cycles

Throughput = Frames x Block Length / Latency = 1 x 12/30 = 0.4 per clock cycles

The code is parameterized, and the simulation has been run for a small H matrix for easy understanding. This also results in a smaller throughput.